



ONLINE FOOD DELIVERY ANALYSIS: DATA DRIVEN BUSINESS INSIGHTS

Data Analytics Project 

Technology Stack: Python | MySQL | Streamlit | Pandas | Plotly | SQLAlchemy

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Overview

The Online Food Delivery Analytics Dashboard is a data analytics project developed to analyze food delivery operations and provide meaningful business insights through interactive visualizations and SQL-based analysis. The project utilizes Python, MySQL, Pandas, Plotly, and Streamlit to process, analyze, and visualize food delivery data.

The dashboard enables users to monitor important business metrics such as total revenue, total orders, average delivery time, customer ratings, profit margins, and restaurant performance. It also provides SQL query execution for business analysis and exploratory data analysis (EDA) using interactive charts.

This project demonstrates the complete data analytics workflow, including data cleaning, feature engineering, database integration, SQL analysis, dashboard development, and business insight generation.

1. INTRODUCTION

The rapid growth of online food delivery services has generated large volumes of customer and operational data. Analyzing this data helps businesses understand customer behavior, optimize delivery operations, improve restaurant performance, and increase profitability.

The objective of this project is to develop an interactive dashboard that transforms raw food delivery data into meaningful business insights. The dashboard allows users to explore data through SQL queries, KPIs, and visualizations, enabling informed business decisions.

2. PROBLEM STATEMENT

Food delivery companies handle thousands of customer orders every day. Without proper analysis, businesses struggle to identify revenue trends, customer preferences, delivery performance, and operational inefficiencies.

The challenge is to develop an interactive analytics dashboard that enables users to analyze food delivery data efficiently and supports better business decision-making.

3. OBJECTIVES

The primary objectives of this project are:

- Analyze food delivery operations
- Monitor business KPIs
- Evaluate restaurant performance
- Study customer behavior

- Analyze customer ratings
 - Measure delivery efficiency
 - Understand the impact of discounts on revenue
 - Perform SQL-based business analysis
 - Develop an interactive Streamlit dashboard
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4. TECHNOLOGY STACK

Programming Language:

- Python

Database:

- MySQL

Libraries:

- Pandas
- NumPy
- Plotly
- Matplotlib
- Seaborn
- SQLAlchemy

Framework:

- Streamlit

Development Environment:

- Visual Studio Code
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5. DATASET DESCRIPTION

The dataset consists of transactional records from an online food delivery platform, containing information about customers, restaurants, orders, payments, deliveries, and business performance. It was used to analyze customer behavior, restaurant performance, revenue generation, delivery efficiency, and profitability.

- **Order ID** – Unique identifier for each order.
- **Customer ID** – Unique identifier for each customer.
- **Restaurant Name** – Name of the restaurant fulfilling the order.
- **City** – City where the order was placed.
- **Cuisine Type** – Type of cuisine ordered (e.g., Indian, Chinese, Italian).
- **Order Value** – Total value of the order before discounts.
- **Discount Applied** – Discount amount offered on the order.
- **Final Amount** – Amount paid by the customer after applying discounts.
- **Profit Margin (%)** – Profit earned on the order expressed as a percentage.
- **Customer Rating** – Rating provided by the customer after delivery.
- **Payment Method** – Mode of payment (UPI, Credit Card, Cash, Wallet, etc.).
- **Order Status** – Status of the order (Delivered, Cancelled, etc.).

The dataset was cleaned and transformed before analysis to ensure accuracy and consistency.

6. DATA CLEANING

The following preprocessing steps were performed :

- Removed duplicate records
- Checked for missing values

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- Converted data types
 - Standardized column names
 - Corrected inconsistent values
 - Verified numerical columns
 - Prepared the dataset for SQL import

These steps improved data quality and ensured reliable analytical results.

7. FEATURE ENGINEERING

Several new features were created to improve analysis.

- Order day type
- Customer age group
- Delivery performance category
- Profit margin percentage

Feature engineering enhanced the dataset by creating meaningful variables for deeper analysis.

8. DATABASE DESIGN

The cleaned dataset was imported into a MySQL database.

The database contains a delivery table storing all transactional records.

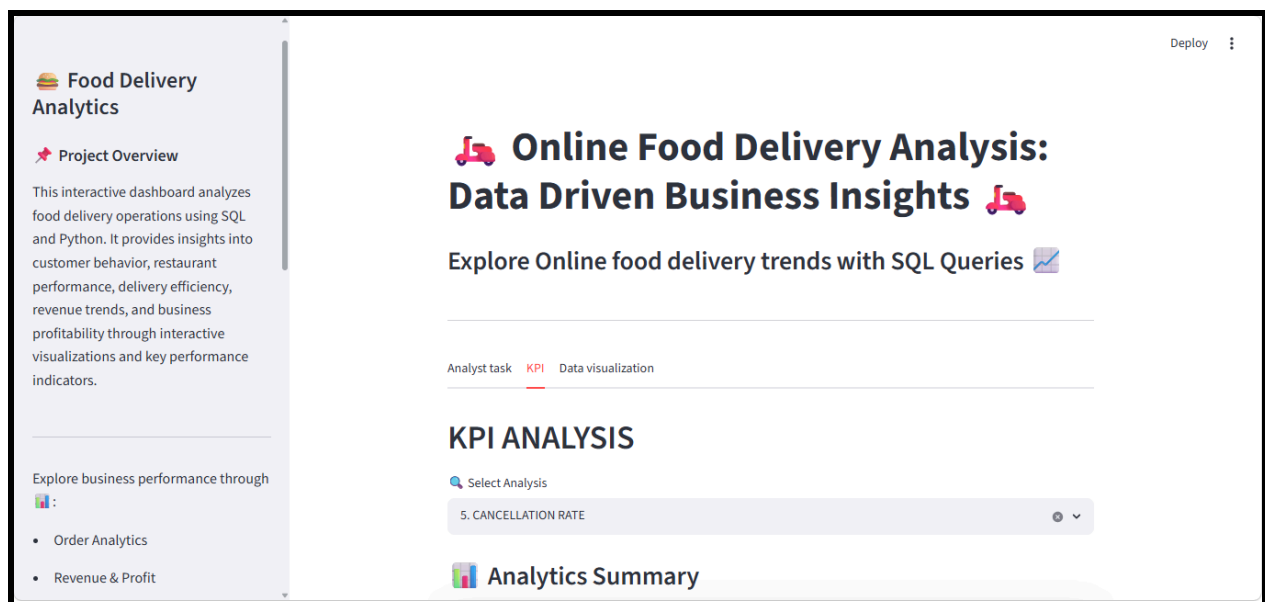
Python was connected to MySQL using SQLAlchemy, enabling efficient execution of SQL queries within the Streamlit application.

9. DASHBOARD DEVELOPMENT

An interactive dashboard was developed using Streamlit.

The dashboard consists of multiple modules.

- KPI Dashboard
- SQL Analysis
- Exploratory Data Analysis
- Side Bar
- Business Insights



10. SQL ANALYSIS

Several SQL queries were developed to answer important business questions.

Examples include:

- Identify Top Spending Customers
- Analyze Age Group vs Order Value

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- Weekend vs Weekday Order Patterns
 - Monthly Revenue Trends
 - Impact of Discounts on Profit
 - High-Revenue Cities and Cuisines
 - Average Delivery Time by City
 - Distance vs Delivery Delay Analysis
 - Delivery Rating vs Delivery Time
 - Top-Rated Restaurants
 - Cancellation Rate by Restaurant
 - Cuisine-wise Performance
 - Peak Hour Demand Analysis
 - Payment Mode Preferences
 - Cancellation Reason Analysis

These queries provide quick access to key business metrics.

11. KEY PERFORMANCE INDICATORS (KPIs)

The dashboard includes several important KPIs.

- Total Orders
- Total Revenue
- Average Order Value
- Average Delivery Time
- Profit Margin percentage
- Average Delivery Rating
- Cancellation Rate

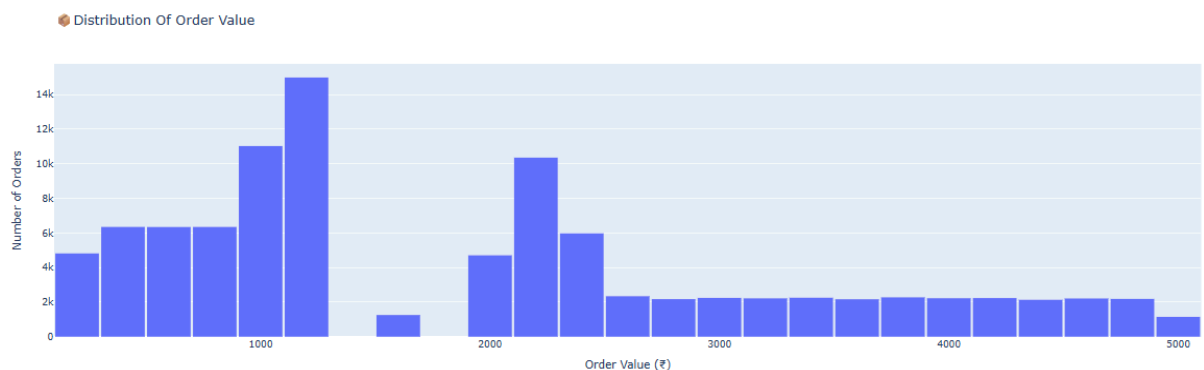
These KPIs provide a quick overview of business performance.

12. Exploratory Data Analysis (EDA)

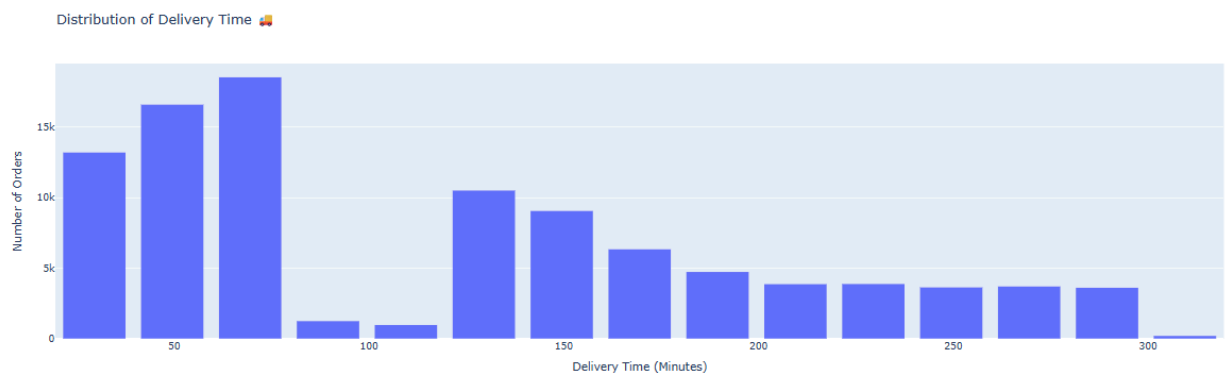
Exploratory Data Analysis (EDA) was performed to understand the Online Food Delivery dataset and identify important business insights. Different visualizations were created using Matplotlib, Seaborn, and Plotly to analyze customer behavior, delivery performance, revenue, and operational trends.

EDA Topics

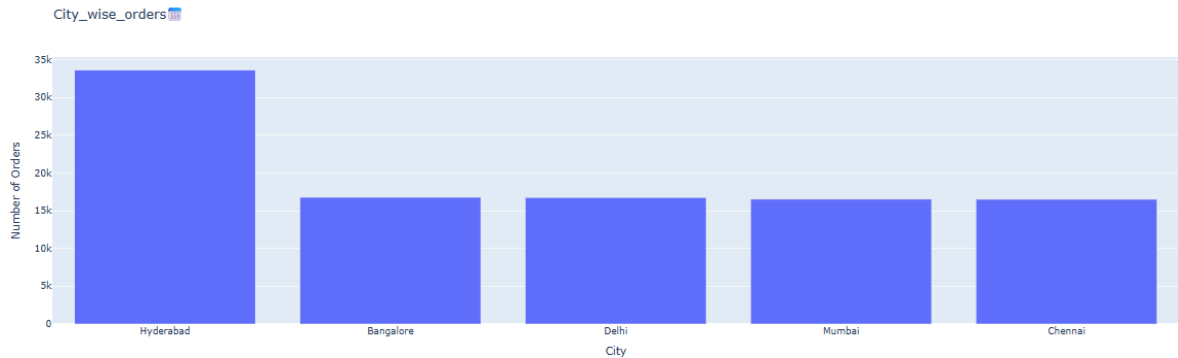
- **Distribution of Order Values** – Analyzed customer spending patterns and the distribution of order values.



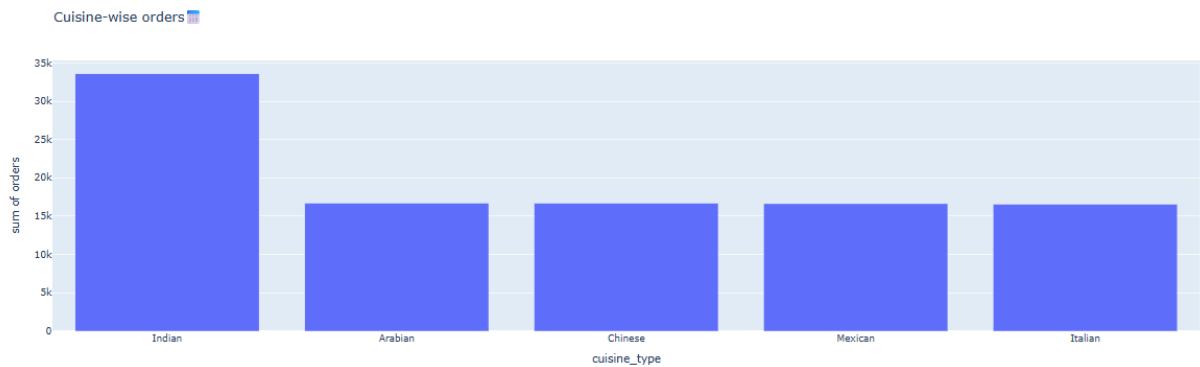
- **Distribution of Delivery Time** – Examined delivery time patterns and identified delivery delays.



- **City-wise Order Analysis** – Compared order count, revenue, and delivery performance across cities.



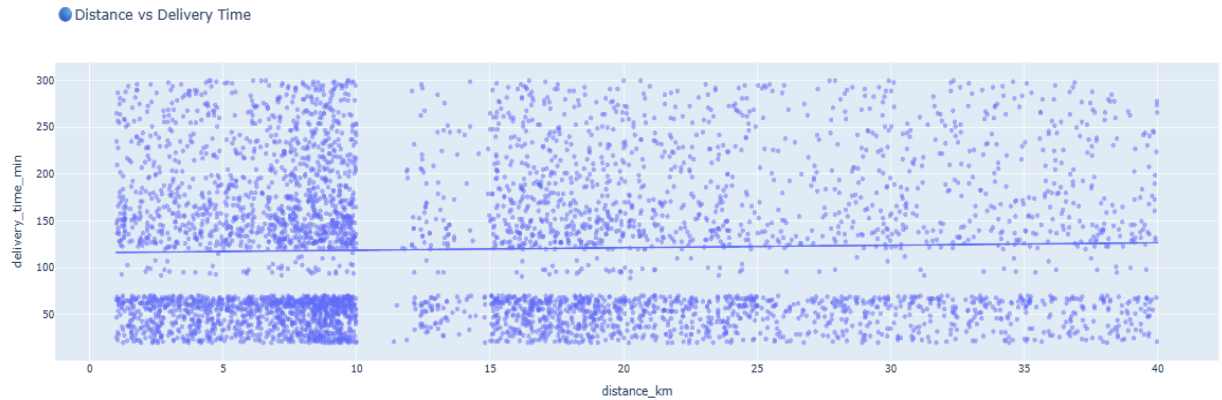
- **Cuisine-wise Order Analysis** – Analyzed the popularity and performance of different cuisine types.



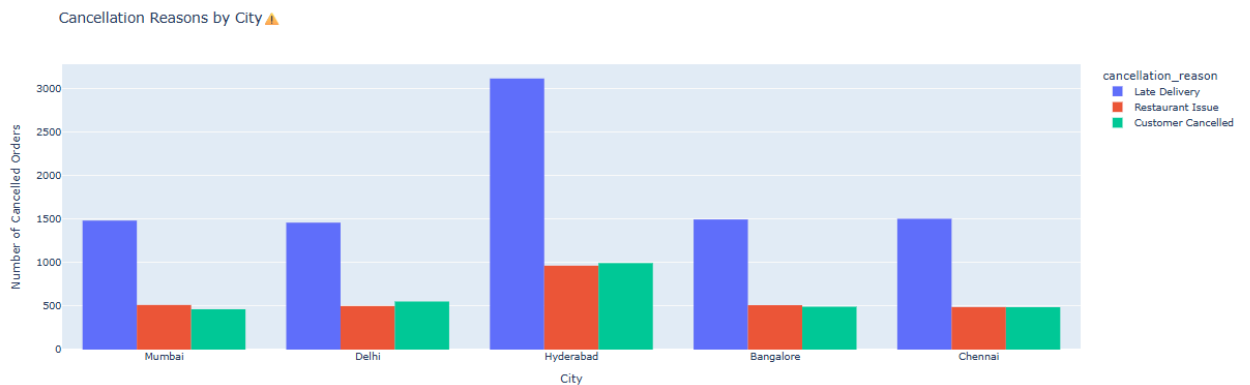
- **Weekend vs Weekday Demand** – Compared customer order patterns between weekdays and weekends.



- **Distance vs Delivery Delay Analysis** – Studied the relationship between delivery distance and delivery time.



- **Cancellation Reason Analysis** – Identified the major reasons for order cancellations.



- **Correlation Analysis Among Numeric Features** – Examined the relationships between numerical variables using a correlation heatmap.



The EDA helped uncover trends, customer preferences, delivery performance, and business patterns. Multiple visualizations under each analysis topic provided valuable insights, which were used to support decision-making and enhance the interactive dashboard.

13. Business Insights

The analysis of the Online Food Delivery dataset revealed several key business insights:

- Cities with higher order volumes generated greater revenue.
- Top-performing restaurants consistently received more orders and higher revenue.
- Restaurants with better customer ratings attracted more customers.
- UPI and other online payment methods were the most preferred payment options.

- Moderate discounts encouraged more customer orders and increased sales.
- Excessive discounts reduced overall profit margins.
- Faster deliveries resulted in higher customer satisfaction and better delivery ratings.
- Longer delivery times were associated with lower customer ratings.
- Delivery distance had a direct impact on delivery time.
- Customer cancellations were mainly driven by a few common reasons, helping identify areas for operational improvement.

These insights can help businesses improve customer experience, optimize delivery operations, increase revenue, and make informed strategic decisions.

14. CHALLENGES FACED

During the project, several challenges were encountered.

- Missing values in the dataset
- Data type conversion issues
- SQL query optimization
- Database connectivity
- Streamlit layout customization
- Visualization formatting

Each challenge was resolved through data preprocessing, debugging, and code optimization.

15. FUTURE ENHANCEMENTS

Future improvements include:

- Machine Learning prediction models
- Delivery time prediction

- Customer segmentation
 - Restaurant recommendation system
 - Real-time dashboard updates
 - Cloud deployment
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16. CONCLUSION

The dashboard enables users to monitor business performance, identify trends, and make informed decisions based on data. It highlights the practical application of Python, SQL, and Streamlit in solving real-world business problems and showcases essential data analytics skills.
